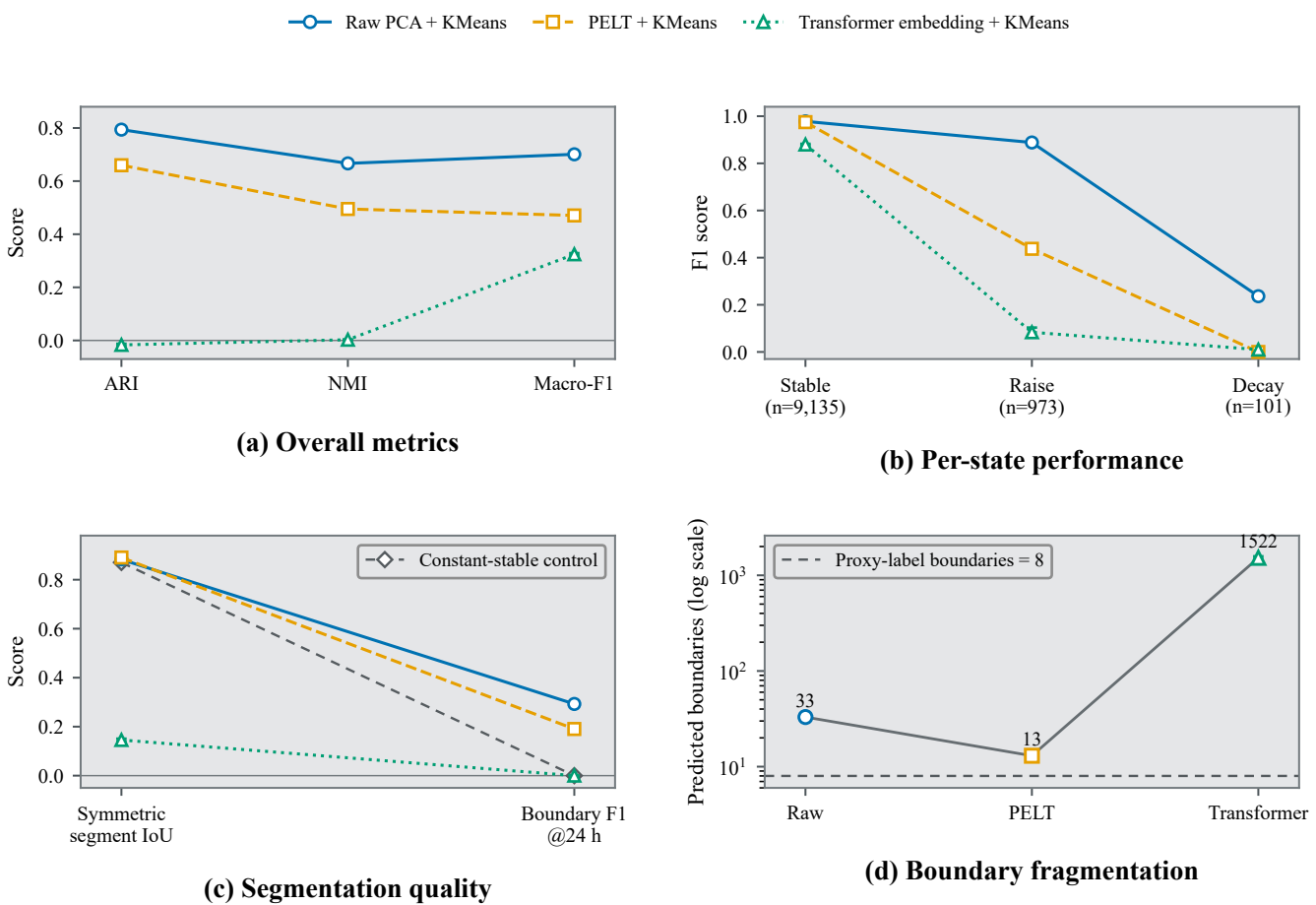

Raw PCA + KMeans
PELT + KMeans
Transformer embedding + KMeans
Score
0.0
0.2
0.4
0.6
0.8
ARI
NMI
Macro-F1
(a) Overall metrics
F1 score
0.0
0.2
0.4
0.6
0.8
1.0
Stable
(n=9,135)
Raise
(n=973)
Decay
(n=101)
(b) Per-state performance
Score
Constant-stable control
Symmetric
segment IoU
Boundary F1
@24 h
(c) Segmentation quality
Predicted boundaries (log scale)
Proxy-label boundaries = 8
33
13
1522
Raw
PELT
Transformer
(d) Boundary fragmentation